

EVEN SEMESTER EXAMINATION, 2023 – 24
2ND yr B.Tech. – CS&E /AI&ML
DIGITAL ELECTRONICS

Duration: 3:00 hrs

Max Marks: 100

Note: - Attempt all questions. All Questions carry equal marks. In case of any ambiguity or missing data, the same may be assumed and state the assumption made in the answer.

Q 1.	Answer any four parts of the following. a) Define the term universal gates and their applications. Define the term universal gates and their applications. b) Convert the following, 1. $(5162)_{10} = ()_2$ 2. $(11011001)_2 = ()_{10}$ 3. $(6273)_{10} = ()_8$ 4. $(7860)_{10} = ()_{16}$ 5. $(A23B8)_{16} = ()_{10}$ c) What are the VHDL constructs used for combinational circuits and sequential circuits? d) Using active high output 3:8 line decoder, implement the following functions $F_1(A, B, C, D) = \sum m(0, 1, 2, 5, 7, 11, 15)$ $F_2(A, B, C, D) = \prod m(1, 3, 4, 11, 13, 14)$ e) Define the TTL (Transistor-Transistor-Logic) logic Family used for digital circuits. f) Define the De-morgan's theorem of Logic Simplification for SOP & POS forms.	5x4=20
Q 2.	Answer any four parts of the following. a) Elaborate the characteristic equations of S-R and J-K Flip-Flops. b) Explain difference between latch and flip-flop. c) What are the differences between Combinational Circuits and Sequential Circuits? Convert the Gray Code 111001 into binary code. d) What is race around condition in JK flip-flop? Find the 'x' base of $(211)_x = (152)_8$ e) Define the following term in digital circuits? i) Noise Margin, ii) Propagation delay, iii) Fan-in, iv) fan-out. f) What is the programmable table of PLA? How is the capacity of a PLA specified?	5x4=20
Q 3.	Answer any two parts of the following. a) Design a 4-bit binary up down ripple counter, also show its clock diagram. b) Simplify the Boolean function $F(A, B, C, D) = \sum (1, 3, 7, 11, 12, 13)$ which has the don't care condition $d(A, B, C, D) = \sum (0, 2, 5, 9)$ and then express the simplified function in sum-of-minterms form. c) What is a TTL NAND gate? Draw the circuit diagram of a standard TTL NAND gate.	10x2=20
Q 4.	Answer any two parts of the following. a) Explain the truth table of the SR, JK, D & T flip-flops. b) Solve the logic function given below, using Quine McClusky minimization technique and realize simplified expression using universal gates. $F(A, B, C, D) = \sum (0, 1, 3, 7, 8, 9, 11, 15)$ c) What is dataflow modeling in VHDL? Explain the behavioral modeling in VHDL? What are the advantages of using dataflow modeling?	10x2=20
Q 5.	Answer any two parts of the following. a) Explain different logic gates families in digital circuits. Write a short note on universal gate. b) Solve the following Boolean functions by using K-Map: $F(a, b, c, d) = \sum (0, 1, 4, 5, 6, 8, 9, 10, 12, 13, 14)$ c) What is an FPGA? Describe the basic architecture of an FPGA. How is an FPGA programmed?	10x2=20